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I declare that in submitting all work for this assessment, I have read, understood, and agree to the content and expectations of the Assessment Declaration.

ROYAL MELBOURNE INSTITUTE OF TECHNOLOGY

SGS CAMPUS

LAW2619 – Legal Consideration of Artificial Intelligence, Big Data, and Blockchain

Assignment 3

Hypothetical scenario/ case study

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EXECUTIVE SUMMARY

Constantly researching and adopting emerging technologies is necessary for innovation in hospitals, however, there is also the involvement of new legal and ethical principles. This report will examine the three most concerning legal issues which are Compliance with HIPAA, Liability & Accountability in digital cyberspace, and the Assessment of AI. Failing to accomplish these issues can result severely in the company's reputation, as well as a huge financial loss. The report format will be:

Heading 1: Introduction: Overview of the report

Heading 2: **Background and Legal Considerations**: Evaluating the current situation of the US healthcare industry and giving recommendations, overview of legal issues

Heading 3: Analysis of Emerging Issues: Mentioning and analyzing the three legal issues

- HIPAA Security Rules Compliance: Analyzing the implication of non-compliant behavior on healthcare providers through the ***In re Anthem, Inc. Data Breach Litigation*** case
- Liability and Accountability: Mentioning the US tort law and conditions of negligence, analyze through the ***In re Anthem, Inc. Data Breach Litigation*** case
- Assessment of AI: Updating with emerging AI legal and ethical considerations act, analyze its implications.

Heading 4: Recommendations and Solutions: Providing frameworks and guidelines for each legal issue to assist the company in addressing them.

Heading 5: Conclusion: Compilation of all the insights from the report

ABBREVIATION

AI: Artificial Intelligence

AI HLEG: High-Level Expert Group on Artificial Intelligence

ePHI: electronic protected health information

HIPAA: Health Insurance Portability and Accountability Act

ML: Machine Learning

NIST: National Institute of Standards and Technology

NIST CSF: National Institute of Standards and Technology's Cybersecurity Framework

NLP: Natural Language Processing

PHI: protected health information

I. INTRODUCTION

As a future Information Security Specialist, I conduct this report to examine and have a thorough grasp of the US healthcare background, hence, suggesting an emerging technology to fulfill the demand of society. Moreover, this research also identifies the three severe legal issues that healthcare providers might face when applying the recommended technologies which are (1) *the challenges when complying with HIPAA Security Rules,* (2) *acknowledging the liability and accountability in AI development and deployment,* and (3) *the troubles of complying with emerging AI Policy Act*, evaluate its implication on healthcare providers. Finally, recommend some potential frameworks to address these issues.

II. BACKGROUND AND LEGAL CONSIDERATIONS

1. INDUSTRY CONTEXT

The US is known as the highest-spending country in the world for healthcare, however, the budget is mostly through tax expenditure and out-of-pocket payments.[1] Moreover, less than 10% of US healthcare resources are allocated in rural areas where 20% American population lives.[2] The rising cost of using these services and the lack of availability of services limit access to healthcare – a component that positively impacts the US's life expectancy.[3][4]

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[1] Jacqueline Howard, 'US Spends Most on Health Care but Has Worst Health Outcomes among High-Income Countries, New Report Finds', *CNN* (31 January 2023) <https://edition.cnn.com/2023/01/31/health/us-health-care-spending-global-perspective/index.html>.

[2] Nicholas C. Coombs, Duncan G.Campbell and James Caringi, 'A Qualitative Study of Rural Healthcare Providers' Views of Social, Cultural, and Programmatic Barriers to Healthcare Access' (2022) 22(1) *BMC health services research* 438

[3] Lisha Hao et al, 'Adequate Access to Healthcare and Added Life Expectancy among Older Adults in China' (2020) 20(1) *BMC Geriatrics* 129

[4] Max Roser, 'Why Is Life Expectancy in the US Lower than in Other Rich Countries?', *Our World in Data* (29 October 2020) <https://ourworldindata.org/us-life-expectancy-low>.

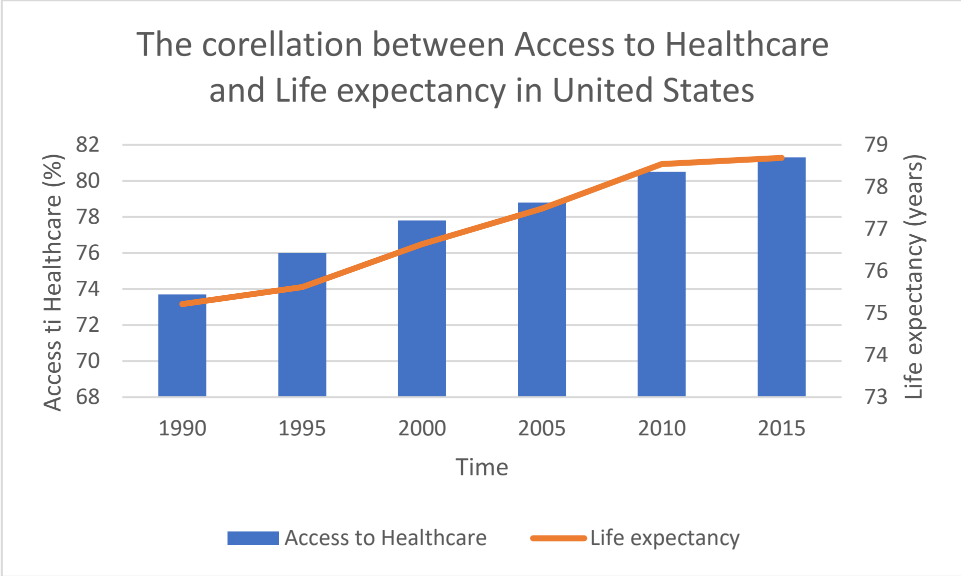

Figure 1[5][6]

Furthermore, with a high percentage of people carrying chronic disease (approximately 45%, almost half of the population) and a discharge rate of around 12% (12.549 discharges/ 100.000 patients), hospitals must strategically monitor the bed admission rate and allocate resources to ensure the hospital's flow rate and manage at-risk patients.[7][8]

2. RECOMMENDATION

Therefore, a recommended step would be to develop and implement an AI-powered chatbot that adopts the supervised and unsupervised ML model. Firstly, it would be integrated with NLP to effectively communicate and answer common-health-related questions and offer medical advice based on the symptoms input by the patients by leveraging the supervised learning aspect, providing personalization, and reducing unnecessary in-person visits.[9]

Secondly, some features would be incorporated such as predictive models using behavioural, genetic, and environmental conditions data by leveraging the unsupervised learning methods; therefore,

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[5] World Bank, 'World Development Indicators', *DataBank* <https://databank.worldbank.org/indicator/SP.DYN.LE00.IN/1ff4a498/Popular-Indicators >.
[6] 'Healthcare Access and Quality Index', *Our World in Data* (7 June 2017) <https://ourworldindata.org/grapher/healthcare-access-and-quality-index?tab=chart>
[7] Wullianallur Raghupathi and Viju Raghupathi, 'An Empirical Study of Chronic Diseases in the United States: A Visual Analytics Approach' (2018) 15(3) *International journal of environmental research and public health* 431
[8] OECD, 'Hospital discharges by diagnostic categories', *OECD* <https://data-explorer.oecd.org/vis?tm=discharge&pg=0&fs%5b0%5d=Measure%2C0%7CHospital%20discharges%23DISCHARGE%23&fc=Measure&snb=3&vw=tb&df%5bds%5d=dsDisseminateFinalDMZ&df%5bid%5d=DSD_HEALTH_PROC%40DF HOSP_DISCHARGE&df%5bag%5d=OECD.ELS.HD&df%5bvs%5d=1.0&dq=USA..DSC_10P5PS...DICDA000......._T>
[9] Shuroug A. Alowais et al, 'Revolutionizing Healthcare: The Role of Artificial Intelligence in Clinical Practice' (2023) 23(1) *BMC Medical Education* 1-689

discovering the pattern of patient lifestyle and alerting if abnormalities exist, help control the readmitted patient rate due to chronic diseases.[10]

Finally, since the US has a low density of hospital beds (approximately 2.42 beds/ 1.000 patients), a real-time monitoring system for bed admission and scheduling doctor appointments is necessary.[11]

3. LEGAL CONSIDERATION

With a vision of becoming an Information Security Specialist, my viewpoint on AI is that this emerging technology can have beneficial impacts on the healthcare industry overload situation in general, however, introducing an AI-powered chatbot in this sensitive industry can raise several legal, regulatory, and ethical challenges.

First and foremost, healthcare providers, especially in the US, must strictly comply with HIPAA to ensure PHI's confidentiality, integrity, and availability and to fulfill the patient's consent and rights. Although the AI-powered system will need robust training data to be effectively put into operation, data privacy and data security must be prioritized.

The second issue would be the liability and accountability in the operation process of AI systems. It raises a huge question about who will be held accountable for explaining the data-related processes, the logic behind the suggested treatment, and liable for the unintended consequences.

Finally, there are rising concerns regarding the algorithm's accuracy, which directly relates to the quality and quantity of data being utilized in AI training. Minimizing algorithmic bias, which is crucial to providing a better health outcome, and preventing the violation of anti-discrimination laws. For instance, if the AI-powered chatbot is being trained with under-presented data, limiting the capability of personalizing the treatment experience for patients with different ethnicities or even interpreting the lack of data into a lack of diseases.[12]

III. ANALYSIS OF EMERGING ISSUES

1. FIRST LEGAL ISSUE

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[10] Ibid.

[11] Jenny Yang, 'Hospital Bed Density by Country', *Statista* (21 February 2024) <https://www.statista.com/statistics/283273/oecd-countries-hospital-bed-density/>.

[12] Malwina Anna Wójcik, 'Algorithmic Discrimination in Health Care: An EU Law Perspective' (2022) 24(1) *Health and human rights* 93

The first legal issue when integrating this AI-powered chatbot is *whether a healthcare provider faces issues complying with HIPAA since it collects PHI to create a personalized experience, posing jurisdictional challenges regarding data security measures*.

Implications

Distinguishing between “data security” and “data privacy” is crucial for legislative compliance. Data security relates to healthcare providers' measures to protect data against unauthorized access.[13] On the contrary, data privacy is the right of an individual to control every personal information-data-related process.[14] Hence, these concepts have given rise to HIPAA Privacy Rules and Security Rules. However, as an Information Security Specialist, I will focus more on the security aspects.

The HIPAA Security Rule is applicable to secure ePHI as it requires the covered entities' measures to protect the confidentiality, availability, and integrity of data.[15]

Any non-compliance behaviour of healthcare providers regarding HIPAA can result in substantial penalties. The penalties can range from 100$ to several thousand dollars per violation[16]. This not only results in severe financial loss but also damages the reputation and trust of patients.[17]

<table>
<tr><th>Penalty Tier</th><th>Culpability</th><th>Minimum-Maximum Penalties/ Violation</th><th>Maximum Annual Penalty</th><th>Note</th></tr>
<tr><td>Tier 1</td><td>Unintentionally/ Unaware Violation</td><td>$100-$50,000 and imprisonment up to 1 year</td><td rowspan="4">$1,500,000</td><td rowspan="4">The penalties haven’t been inflation-adjusted</td></tr>
<tr><td>Tier 2</td><td>Reasonable Causes</td><td>$1,000-$50,000 and imprisonment up to 5 years</td></tr>
<tr><td>Tier 3</td><td>Willful negligence (attempt to correct)</td><td>$10,000-$50,000 and imprisonment up to 10 years</td></tr>
<tr><td>Tier 4</td><td>Willful negligence (no attempt to correct)</td><td>$50,000 and imprisonment up to 10 years</td></tr>
</table>

[13] Richard May and Kerstin Denecke, ‘Security, Privacy, and Healthcare-Related Conversational Agents: A Scoping Review’ (2021) 47(2) *Informatics for Health and Social Care* 194
[14] Ibid.
[15] Health Insurance Portability and Accountability Act (HIPAA) 1996 (US)
[16] American Medical Association, ‘HIPAA Violations & Enforcement’, *American Medical Association* (6 December 2019) <https://www.ama-assn.org/practice-management/hipaa/hipaa-violations-enforcement>.
[17] Steve Alder, ‘Editorial: The Cost of Non-Compliance with HIPAA’, *HIPAA Journal* (20 December 2023) <https://www.hipaajournal.com/cost-non-compliance-hipaa/>.

Figure 2: HIPAA Violation's Penalties[18][19]

Through the ***In re Anthem, Inc. Data Breach Litigation*** case, one of the largest breaches of 79 million people was caused by cyberattacks. Anthem has consolidated the class-action lawsuit for all the data breach victims of $115 million.[20][21] However, they must suffer from the HIPAA violation penalty of $16 million for Civil Rights and substantial penalties for poor security systems that do not meet the standard of HIPAA Security Rules, consisting of violating the minimum requirement of access control § 164.312(a), failing to conduct risk analysis § 164.308(a)(1) …[22][23][24]

2. SECOND LEGAL ISSUE

Directly linked with the first legal issue, the second legal issue is *whether the healthcare providers may face challenges in identifying the responsibility and accountability under HIPAA to hold the entities liable as the system relied on the input data, processed data, and prompt.*

Taking the same ***In re Anthem, Inc. Data Breach Litigation*** case in consideration of US tort law, Anthem has been implied for their negligence as they have not met the duty of care in which they must protect the PHI.[25][26] Due to the inefficient access control and risk analysis, this has breached their duty, causing vulnerability in the system, and facilitating cyberattacks.[27] This resulted in the loss of 79 million patient data.[28]

3. THIRD LEGAL ISSUE

Finally, the third legal issue involved encompassing AI-powered chatbots which is *whether healthcare providers having trouble complying with the emerging AI Policy Act when using AI-powered chatbots to analyze personal data, predict the pattern, and give personalized medical recommendations.*

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[18] Steve Alder, 'What Are the Penalties for HIPAA Violations? 2024 Update', *HIPAA Journal* (1 February 2024) <https://www.hipaajournal.com/what-are-the-penalties-for-hipaa-violations-7096/>.
[19] American Medical Association, 'HIPAA Violations & Enforcement', *American Medical Association* (6 December 2019) <https://www.ama-assn.org/practice-management/hipaa/hipaa-violations-enforcement>.
[20] Kyle Chin, 'Top 20 Worst HIPAA Violation Cases in History', *UpGuard* (8 September 2023) <https://www.upguard.com/blog/worst-hipaa-violation-cases>.
[21] *In re: Anthem, Inc. Data Breach Litigation* (2016) 162 F Supp 3d 953.
[22] Ibid.
[23] Health Insurance Portability and Accountability Act (HIPAA) 1996 (US) § 164.312(a)
[24] Health Insurance Portability and Accountability Act (HIPAA) 1996 (US) § 164.308(a)(1)
[25] Kyle Chin, above n 18
[26] In re: Anthem, Inc. Data Breach Litigation (2016), above n 20.
[27] Kyle Chin, 'Top 20 Worst HIPAA Violation Cases in History', *UpGuard* (8 September 2023) <https://www.upguard.com/blog/worst-hipaa-violation-cases>.
[28] Ibid.

After the endorsement of the EU's AI Act as it represents significant progress in developing a legal framework for these new technologies, many AI-related legislations have been raised to Congress to manage AI's growth and risks.[29] Moreover, Utah's AI Policy Act recently came into effect on 1st May 2024 after passing the bill on 13th March of the same year.[30]

Implications

Although there is a lack of existing AI-related legislation, businesses must still consider legal obligations as there could be failure in the system resulting in biased outcomes.[31] Failing to comply with the EU's AI Act can lead to a 15-35 million Euro financial loss or a 3%-7% loss in the company's global turnover, whichever is higher.[32] Or Utah's AI Policy Act non-compliance can result in $2,5000-5,000/ violation.[33]

IV. RECOMMENDATIONS AND SOLUTIONS

1. FOR FIRST LEGAL ISSUE

The growth of technological threats is growing simultaneously with the rapid expansion of the digital and cyberspace landscape. Every data-related processing has a risk of vulnerability to cybercrime, including unauthorized access and misuse of data.

Hence, HIPAA Security Rules, the division of HIPAA can be found at 45 CFR Part 160 and Part 164 (Subparts A and C), are obligated for entities including health plans, healthcare, Medicare prescription drug card providers, etc.... to comply.[34] However, guidance is absent, making it difficult for businesses to proactively comply with.[35]

[29] Sofia Gracias, 'Comparing the EU AI Act to Proposed AI-Related Legislation in the US', *The University of Chicago Business Law Review* <https://businesslawreview.uchicago.edu/print-archive/comparing-eu-ai-act-proposed-ai-related-legislation-us>.

[30] DataGuidance, 'Utah: AI Policy Act Enters into Force', *DataGuidance* (May 1, 2024) <https://www.dataguidance.com/news/utah-ai-policy-act-enters-force>.

[31] Kenvin D.Pomfret, 'AI Regulation on the Move: EU Leads with AI Act as U.S. States Forge Their Own Paths', *Williams Mullen* (19 March 2024) <https://www.williamsmullen.com/insights/news/legal-news/ai-regulation-move-eu-leads-ai-act-us-states-forge-their-own-paths>.

[32] Blackman Reid and Vasiliu-Feltes, 'The EU's AI Act and How Companies Can Achieve Compliance', *Harvard Business Review* (22 February 2024) <https://hbr.org/2024/02/the-eus-ai-act-and-how-companies-can-achieve-compliance>.

[33] Scott F. Young and Jordan C. Hilton, "Utah Enacts AI-Focused Consumer Protection Bill," *Mayer Brown* (13 May 2024) <https://www.mayerbrown.com/en/insights/publications/2024/05/utah-enacts-ai-focused-consumer-protection-bill>.

[34] HIPAA, 'The Security Rule,' *HHS.gov* (10 September 2009) <https://www.hhs.gov/hipaa/for-professionals/security/index.html>.

[35] Derek Mohammed, 'U.S. Healthcare Industry: Cybersecurity Regulatory and Compliance Issues' (2017) 9(5) *Journal of Research in Business, Economics and Management*

Therefore, the National Institute of Standards and Technology (NIST) has enhanced data security measures through NIST SP 800-66r2, which proposes HIPAA Security Rules compliance guidelines.

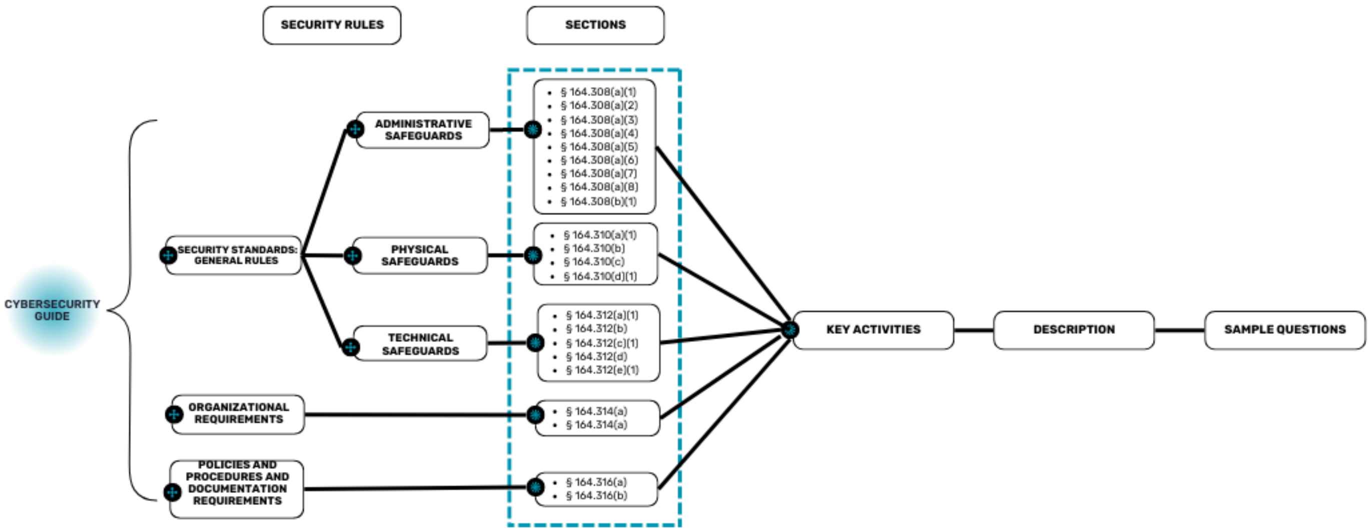

Figure 3: Guidelines for complying with HIPAA Security Rule[36]

Example: HIPAA Security Rules Compliance: § 164.312 (a)(1): *Standard: Access Control*			
Key activities	**Description**	**Sample Questions**	**Implementation Examples**
Assess the workload and operation of each individual	- Identify an approach for access control. - Available for only authorized access, process, and services	- What system contains ePHI? - What application contains ePHI that should only be available for what authorized account?	- Utilizing Role-based access control (RBAC) - Restrict access to the minimum necessary (applying zero-trust architectures)
Grant users with accessibility through different unique user identification	- Assign unique accounts for different users. - Granting accessibility based on activities and responsibilities	- How should the identifier be established? - Are these accesses documented on the log system?	- Requiring multifactor authentication - Configure log generators to document all the operations of users

[36] Jeffrey Marron, 'SP 800-66 Rev. 2, Implementing the Health Insurance Portability and Accountability Act (HIPAA) Security Rule: A Cybersecurity Resource Guide', *CSRC* (14 February 2024) <https://csrc.nist.gov/pubs/sp/800/66/r2/final>.

Figure 3: Applying guidelines for complying with HIPAA Security Rules[37][38][39]

2. FOR SECOND LEGAL ISSUE

To claim entities as negligence under US tort law, it is needed to prove the duty of care, breach of that duty, causation, and damage made by the entities.[40] The NIST CSF 2.0 can facilitate the process of identifying and claiming negligence.

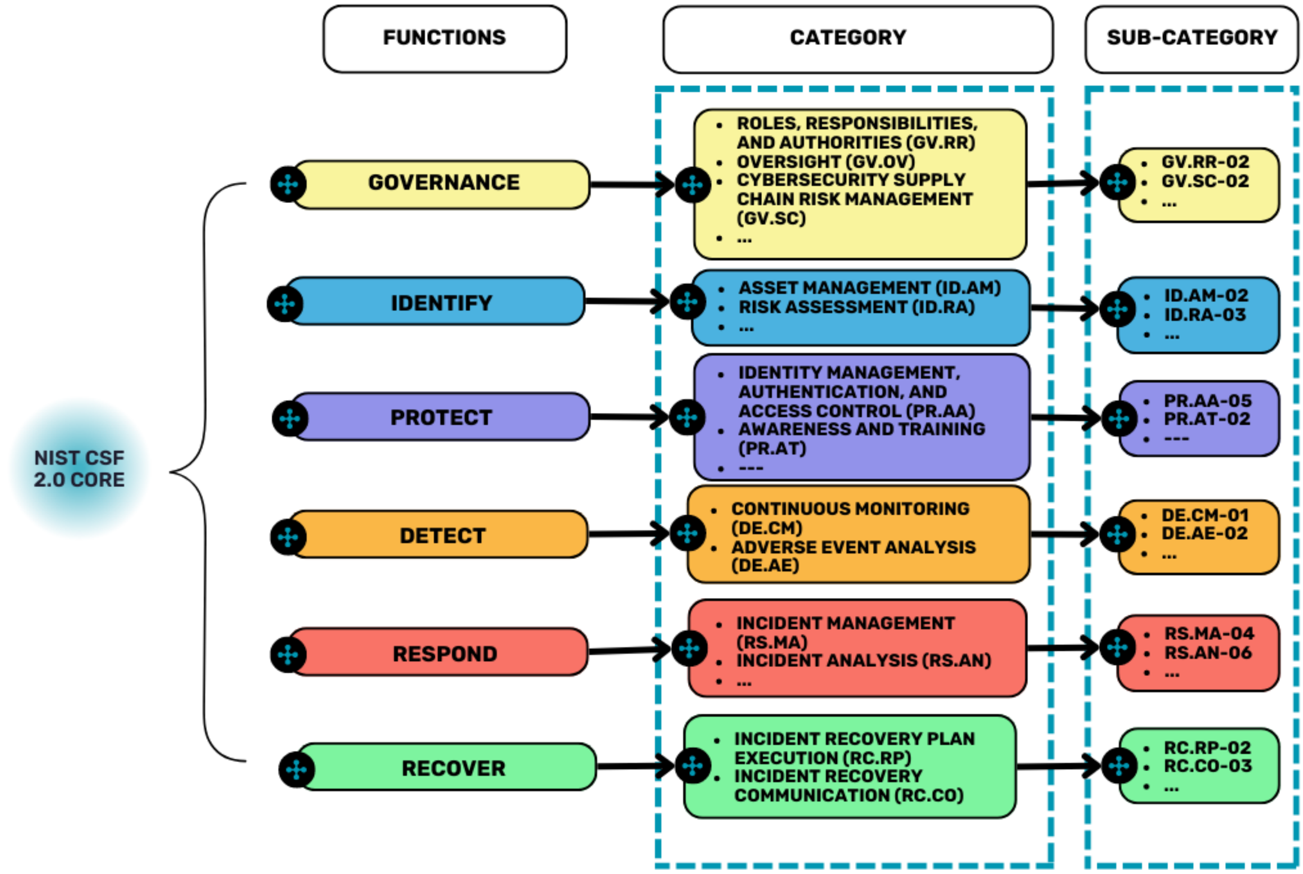

Figure 4: NIST CSF 2.0 core[41]

FUNCTION	CATEGORY	SUB-CATEGORY	ACTIONS

[37] Jeffrey Marron, 'SP 800-66 Rev. 2, Implementing the Health Insurance Portability and Accountability Act (HIPAA) Security Rule: A Cybersecurity Resource Guide', *CSRC* (14 February 2024) <https://csrc.nist.gov/pubs/sp/800/66/r2/final>.
[38] National Institute of Standards and Technology (NIST), 'NIST CSF 2.0 Implementation Examples', *NIST* (26 February 2024) < https://www.nist.gov/system/files/documents/2024/02/21/CSF%202.0%20Implementation%20Examples.pdf>.
[39] Health Insurance Portability and Accountability Act (HIPAA) 1996 (US) § 164.312(a)(1)
[40] LII, 'Tort', *LII / Legal Information Institute* <https://www.law.cornell.edu/wex/tort>.
[41] National Institute of Standards and Technology (NIST), 'The NIST Cybersecurity Framework (CSF) 2.0', *NIST* (26 February 2024) <https://nvlpubs.nist.gov/nistpubs/CSWP/NIST.CSWP.29.pdf>.

GOVERNANCE	Cybersecurity Supply Chain Risk Management (GV.SC)	GV.SC-08: Relevant third-party are included in the incident response plan	Documenting the roles and responsibilities of the organization
PROTECT	Identity Management, Authentication, and Access Control (PR.AA)	PR.AA-05: Privilege and accessibility are defined, managed, and reviewed	Review logical and physical access of authorized entities periodically

Figure 5: NIST CSF's aspects that facilitate identifying negligence and implementation examples.[42][43]

3. FOR THIRD LEGAL ISSUE

To address algorithmic bias and mitigate risks, the utilization of 'Trustworthy AI' is necessary. However, developing a trustworthy and explainable AI (XAI) is still in the early stages in the European Union and not well-known in the US.[44] The term 'Trustworthy AI' appeared on the *Ethics Guidelines for Trustworthy AI* and was coined by the AI HLEG.[45]

The advancement of AI must go with the development of ethical frameworks. Although there were already 100 ethical guidelines, there is still a mismatch when putting these guidelines into practice in AI development, partly due to their lack of practicality.[46] Alleviating these limitations and characterized by *Ethics Guidelines for Trustworthy AI* proposal, Z-Inspection® has employed trustworthiness assessment in every stage of its process lifecycle.[47]

[42] National Institute of Standards and Technology (NIST), 'NIST CSF 2.0 Implementation Examples', *NIST* (26 February 2024) < https://www.nist.gov/system/files/documents/2024/02/21/CSF%202.0%20Implementation%20Examples.pdf>.
[43] National Institute of Standards and Technology (NIST), above n 40
[44] A.S. Albahri, Ali M. Duhaim, Mohammed A. Fadhel, Alhamzah Alnoor, Noor S. Baqer, Laith Alzubaidi, O.S. Albahri, A.H. Alamoodi, Jinshuai Bai, Asma Salhi, Jose Santamaría, Chun Ouyang, Ashish Gupta, Yuantong Gu, Muhammet Deveci, 'A Systematic Review of Trustworthy and Explainable Artificial Intelligence in Healthcare: Assessment of Quality, Bias Risk, and Data Fusion' (2023) 96 *Information Fusion* 156-191
[45] European Commission, 'Ethics Guidelines for Trustworthy AI', *Europa* (8 April 2019) <https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai>.
[46] Vetter, Dennis et al., 'Lessons Learned from Assessing Trustworthy AI in Practice' (2023) 2(3) *Digital Society* 35.
[47] Ibid.

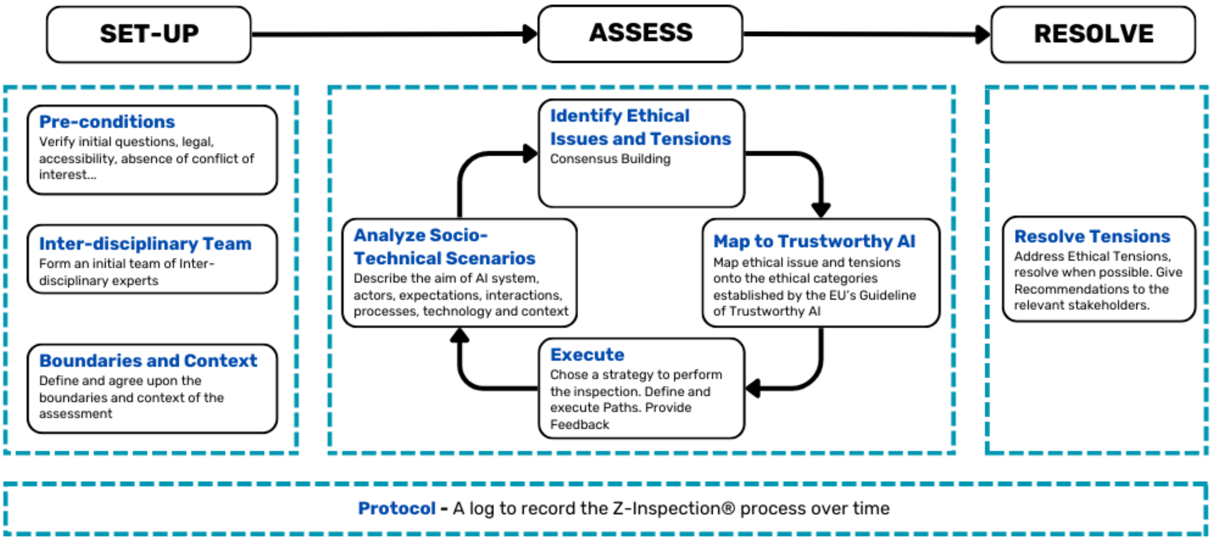

Figure 6: Z-Inspection®'s process flow chart[48]

Hypothetical context: If there is an algorithmic bias in the AI-powered chatbot, failing to make the correct medical recommendation to people with and without chronic diseases

Phase	**Steps**	**Actions**	**Note**
Set-Up	Pre-conditions	Assuring the AI algorithms comply with anti-discrimination laws and the absence of conflict of interest	
	Inter-disciplinary team	Forming a team of 3 Domain experts (doctors…), 2 Ethicists, 2 Legal Experts, 3 Technical experts (AI specialists, data auditors…), 2 Use-case owners	An assessment team should consist of 10-20 interdisciplinary experts
	Boundaries and Context	Using historical medical recommendation data, evaluating the difference in recommendations between people who have and don't have chronic diseases	

48 Vetter, Dennis et al., 'Lessons Learned from Assessing Trustworthy AI in Practice' (2023) 2(3) *Digital Society* 35.

Assess	Analyze Socio-Technical Scenarios	Suggesting the same treatment for both people, making his/her chronic disease relapse and create fatigue.	
	Identify Ethical Issues and Tensions	Not adapting to the chronic health status/ Personalization vs. solidarity	
	Map to Trustworthy AI	**Respect for human autonomy** → Diversity, non-discrimination, and fairness: personalizing recommendations can boost patient autonomy in deciding treatment solutions.	Asking experts to map to the four ethics principles, seven requirements of ethical AI, and sub-requirements to be specific to the problems
	Execute	Assessing the resources to conduct the execution plan	
Resolve	Resolve Tensions	Conducting regular audits on algorithms, mitigating risk through constantly updating the latest data	

Figure 7: Applying the case to Z-Inspection® Framework[49][50]

V. CONCLUSION

Consequently, this report has covered three different guidelines for three legal issues dedicated to healthcare providers. Regarding the challenges in complying with HIPAA Security Rules to prevent issues in security measures, the NIST SP 800-66r2 has been suggested for designing security measures appropriate for a HIPAA Security Rules compliance system. Moreover, implementing the NIST CSF 2.0 helps healthcare providers assess the liability and accountability of an entity for claiming if the

[49] Vetter, Dennis et al., 'Lessons Learned from Assessing Trustworthy AI in Practice' (2023) 2(3) *Digital Society* 35.

[50] Zicari, Roberto V et al., 'Z-Inspection: A Process to Assess Trustworthy AI' (2021) 2(2) *IEEE Transactions on Technology and Society* 83.

performance was negligent. Finally, the Z-Inspection® Framework assists healthcare providers in assessing the AI-powered chatbot for their trustworthiness to prevent the algorithmic bias that creates discrimination.

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